Smart Constructors

Season: 6 · FP Foundations · Category: Functional · Level: Intermediate

Intent

- Make illegal states unrepresentable.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Turnstiles that only admit valid tickets.

When to Use

- Pure functional style in Java 25 + Zazr.
- The design force named in Intent shows up more than once.

When Not to Use

- Throwing exceptions everywhere is 'fine' for a script.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.adt;

import dev.zazr.control.Either;

/** Smart constructor — prefer {@link #parse}; canonical ctor still validates. */
public record Email(String value) {
    public Email {
        if (value == null || !value.contains("@")) {
            throw new IllegalArgumentException("invalid email");
        }
    }

    public static Either<String, Email> parse(String raw) {
        if (raw == null || !raw.contains("@")) return Either.left("invalid email");
        return Either.right(new Email(raw));
    }
}
```

Modern Java 25 / Functional

- Primary examples use Zazr; VAVR shown in comparison episode.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/adt/Email
```

Pitfalls

- Zazr rejects null inside Option/collections.
- Keep interpreters at the edge.

Interview Q&A

- Q: Interview: what problem does Smart Constructors solve? A: Make illegal states unrepresentable.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Make illegal states unrepresentable.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Turnstiles that only admit valid tickets. [0:40] Intent — State the design force in one sentence: Ma...